BSCS FINAL YEAR PROJECT
SRS PHASE-1

SPORTS ECOSYSTEM PLATFORM

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Software Requirements Specification

Version <Version>

SPORTS ECOSYSTEM PLATFORM

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Revision History

Name	Date	Reason For Changes	Version

Previous Phases Feedback

Idea Defence Feedback (Screenshot)

Abstract

The Sports Ecosystem Platform is a web-based-software application inspired by the identifiable gap in the industry when it comes to having a single digital platform to deal with sports training, coaching services, indoor ground bookings, and sales of sports equipment. As it stands, the sports related business, players and coaches all operate independently and therefore there is poor communication, little access to certified coach, manual scheduling and no performance tracking. The proposed project suggests addressing these problems with the help of an organized, centralized system that will help to interconnect all the stakeholders in a single digital environment.

The platform allows players to create profiles, become members, automatically get recommendations of coaches based on the kind of sport, level of skills, and whereabouts, reserve indoor grounds, and monitor training schedules, attendance and performance points. Coaches have the ability to administer their profiles, add verification files, set their availability, accept training invitations, create performance appraisal, and track the players. Business owners receive applications to build and operate online stores of sports equipment, manage the stock of products and orders. Admin panel also provides system integrity by verifying, monitoring and in general management of the platform.Such areas of knowledge as software engineering, web application development, database design, requirement analysis, and system modeling are used in this project. Its application hinges on a role-based access control, secure access, and effective data management as measures to make it reliable and scalable.

It is expected that by the end of the project, an operational sports management ecosystem will be realized and make sports training and services more accessible, transparent and efficient. The system will facilitate a better level of data-driven decision-making, improved user experience and enable a viable digital solution to the current sports management.

1. Introduction and Background

The fast advancement of digital developments in technology have modified the way services are delivered across various industries, including sports and fitness. Most conventional sports management systems are usually based on manual operation, personal recommendation and non-joinable communication lines. Consequently, players often have problems when seeking qualified coaches, reservation of indoor fields, training times, monitoring performance development, and acquiring genuine sportswear in good sources. These are some of the struggles that have led to a collective need to adopt integrated and consolidated computerized solution that might streamline sports related activities within one integrated solution.

Despite the increasing popularity of sports such as cricket and badminton, there is still a lack of structured systems that connect all stakeholders in one ecosystem. Players often depend on informal networks to locate coaches, while coaches struggle to reach potential trainees efficiently. Similarly, business owners selling sports equipment operate independently without direct integration with training or booking services. This fragmentation results in inefficiencies, lack of transparency, time wastage, and limited growth opportunities for stakeholders.

To overcome such problems this project will design of Sports Ecosystem Platform, a centralized web-based system that connects players, coaches, business owners, and administrators into one integrated environment. The platform is designed to digitize and automate key sports-related operations including coach recommendations, indoor ground booking, training session management, attendance tracking, performance evaluation, feedback management, and online equipment sales.

The system adds intelligent services like automated coach recommendation by profile of players, skill and location. It also enables to organize the training management because coaches can impose performance points, track attendance, and the performance of players over time. Moreover, the inclusion of an online sports equipment store in the same platform makes it convenient and accessible to the user, as he would not have to use several different services.

Administratively, the website is transparent and secure because it has user verification, document approval, complaint management, and monitoring. This increases credibility and confidence in the ecosystem. The system creates efficiency in the overall sports activities, communication between the parties, and development of professional sport actively by centralizing all sports-related services.

1.1 Product (Problem Statement)

The Sports Ecosystem represents a software-based web platform offered to offer an inclusive platform through which it is possible to manage sports training, coaching, indoor ground bookings, and sports equipment sales. Players are currently using a manual or unconnected platform to locate coaches, organize training, performance, and acquire equipment. This inefficiency, poor coordination and unmonitored limits are caused by the lack of integration.

The proposed system will eliminate this issue by providing the single solution, under which players can be registered, automatically recommended coaches, training sessions and book of indoor grounds, to track their performance, under the administrative supervision of they coach and business owners of their services respectively.

1.2 Background

The sports training industry implies a number of stakeholders such as players, coaches, facilities providers, and salesmen of equipment. Historically, these stakeholders have been functioning in isolation and hence coordinating them is not easy. As the other organised training, performance tracking, and systematic use of facilities demand more and more structure, centralized digital platform is required.

The field in which this project functions is sports management systems, mainly on indoor sports, e.g. cricket and badminton. This system provides a training management, booking of a facility and e-commerce facility as one ecosystem to assist service providers as well as the users.

1.3 Scope

The area of this SRS will include the design and functional necessities of the Sports Ecosystem system. The system includes:

- Registration and coach suggestions of players and their profile.
- The registration, verification and training of coaches.

Features: Indoor sports Booking of cricket and badminton ground.

- Planning and tracking of training sessions.

Performance appraisal and training plans once a week.

- Management by business owners of sports equipment stores.
- Monitoring, reporting and administrative verification.

1.4 Objectives

The goals of the project are:

- To offer a sports training and management centralized platform.
- To make coach recommendations an automated process that depends on the profile of the player and the location.
- To allow the successful reservation of sports grounds indoors.
- To monitor attendance and performance of the players on a digital platform.
- To enable business owners to sell sports equipments in the internet.
- To provide system reliability in terms of administrative verification and monitoring.

1.5 Challenges

The main obstacles that are discussed within the framework of the present project are:

- The use of various user permissions.
- Providing correct coach recommendations on the basis of player data.
- Organizing the training and ground availability.
- Sustaining data security and integrity.
- Reactions to real time files and updates.

1.6 Learning Outcomes

The following learning outcomes have been attained after working on this project:
Precision of software requirement study and report.

- Experience in multi-role system design.

The understanding of CRUD, workflow management.

- Better awareness of system security and control of access.
- Scalability record in relations to designing sports management systems.

1.7 Nature of End Product

The final product is an online Sports Ecosystem system that allows players, coaches, entrepreneurs and administrators to access it using their own roles. Through a built in system, the system assists in training management, service bookings, performance monitoring as well as equipment sales online.

1.8 Completeness Criteria

The project will be termed as complete when:

- Every user is able to register a user role and log in successfully.

Coach suggestions work as expected.

- There are no conflicts in training sessions, and in ground bookings.
- There is accuracy in recording performance points and attendance.
- Owners of businesses are able to control products and orders.
- Features of admin verification and monitoring are completely active.

1.9 Business Goals

The business objectives that are tackled in this software are:

- Enhancing the avenue of professional sports coaching.
- Vancommoding the use of indoor sporting facilities.
- Offering an online shop of sport supplies.
- Increasing transparency and accountability with performance following.
- Scalability support to support future sport category.

1.10 Related Work/ Literature Survey/ Literature Review

Current management systems in sports are usually designed with one of the booking aspects of facilities or training allocation. Not many services incorporate coaching, monitoring of performances, ground booking, and resale of equipment in one system. The project is based on these ideas and incorporates more than one service into a single ecosystem that enhances the efficiency and the user experience.

1.11 Document Conventions

This Software Requirements Specification is written in a structured format in numeric sections with subsections.

- Headings are bold, and the numbers are made clear.

Functional words like Player, Coach, Admin, and Business Owner are capitalized to depict system roles.

Diagrams used further in the paper are based on standard UML notation.

2. Overall Description

2.1 Product Features

Significant characteristics for the Sports Ecosystem Platform consist of:

- Registration and safe logging in of the players.
- Coach registration, verification and availability administration.
- Self-service coach recommendation system.
- Request and schedule of training sessions.

Indoor ground booking system:

- The tracking of attendance and performance.
- Today, sport equipment retailing on the Web.
- Admin checking and surveillance system.
- Reminder and update notification system.
- Player Payments to Coaches: Players are able to pay coaches on the platform through secure forms of card payment. Major debit/credit cards will be accepted in the system, and the records will be able to maintain the accountability of the transactions.

Business Owner Listing Payments: Companies may pay to have their products listed on the site:

- Entry Level: 20 listings a month.
- Pro Pack - 40 listings in a month.

Premium Package: 60 listings every month.

- Every subscription will be billed on secure card payments, and automated reminders of renewal and generation of invoices.

2.2 User Classes and Characteristics

- Player (End User): This is the person, which is non technical, trains or books in a system, enters the system with the assistance of web and mobile.
- Coach: semi-technical user, controls training schedules and attendance, allocates performance points.

Basic technical skills, products, orders, inventory, and pricing, Business Owner.

- Admin: Intense user, does system verification, system monitoring, can access complete system.

2.3 Operating Environment

The platform shall be web-based.

- Browsers (Chrome, Edge, Firefox).
- Back end server and central database will be employed.
- Win and Linux server compatible.

2.4 Design and Implementation Constraints

- The role-based access control will be adhered to.
- There will be a need of secure authentication.
- There will be a need to ensure that the database remains consistent.
- The system will be limited to Cricket and Badminton only.
- Internet connectivity will be necessary.
- Internet connectivity will be required.

2.5 Assumptions and Dependencies

- Customers will be able to access the internet.

- Coaches and owners of businesses will present valid documents.
- Verification will be done by the system administrator.
- Database server will be available round the clock.
- It will support browser compatibility.

3. Functional Requirements

The Sports Ecosystem Platform has all its functional requirements in the form of use cases defined. The use cases relate to an interaction between the system and external actor. These use cases together define how the system behaves in different scenarios

Use Case Diagram (Overview)

The key players of the system are the following:

- Player
- Coach
- Business Owner
- Admin

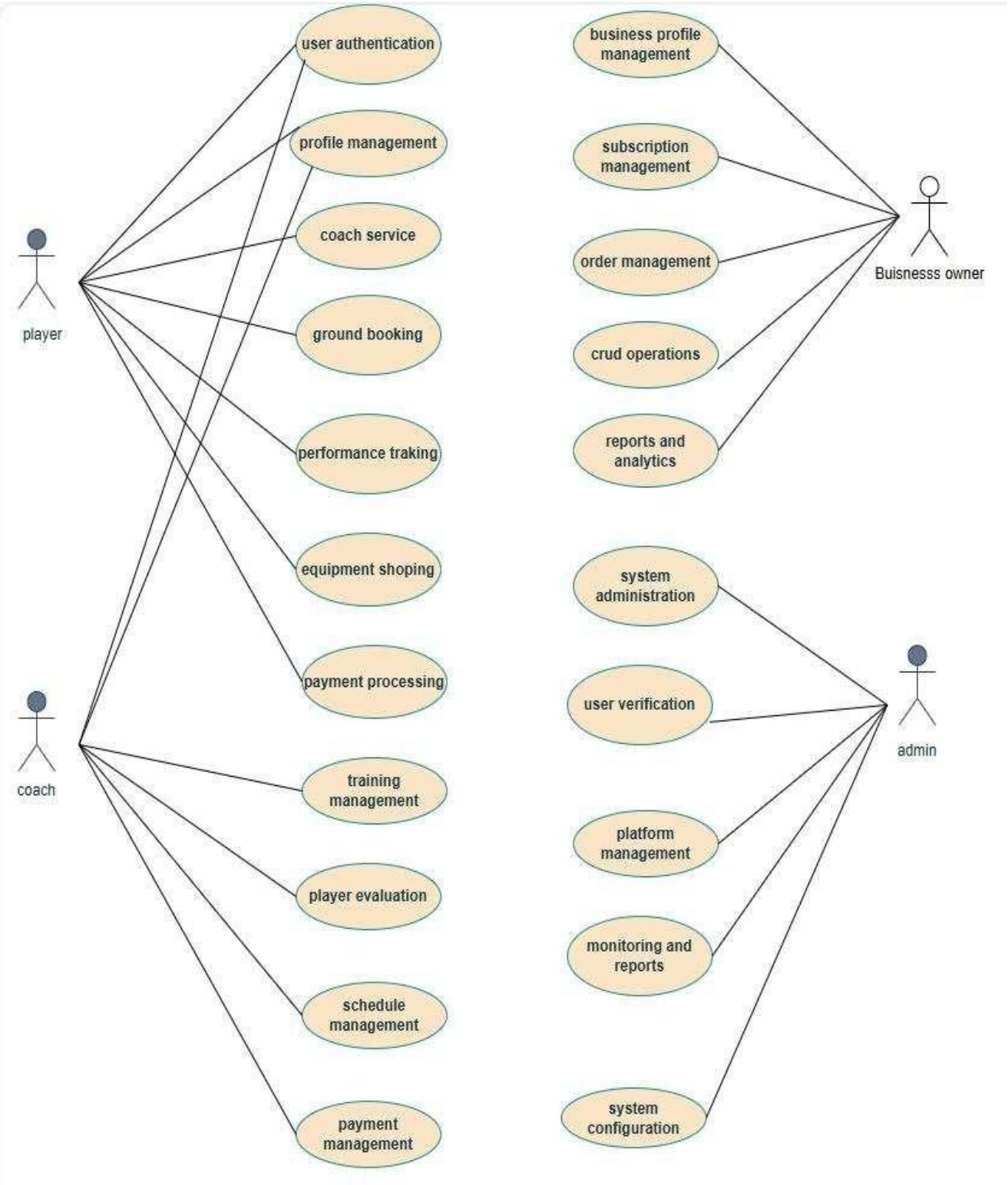
The systems handle the Player, who communicates with the system in order to register, choose the coaches, order the trainings, and reserve the indoor fields, order the sports equipment.

The Coach handles the training sessions, as well as attendance and player performance.

Sports products and orders are handled by the Business Owner.

The Admin manages the verification, monitoring, and the general management of the system.

USE CASE DIAGRAM



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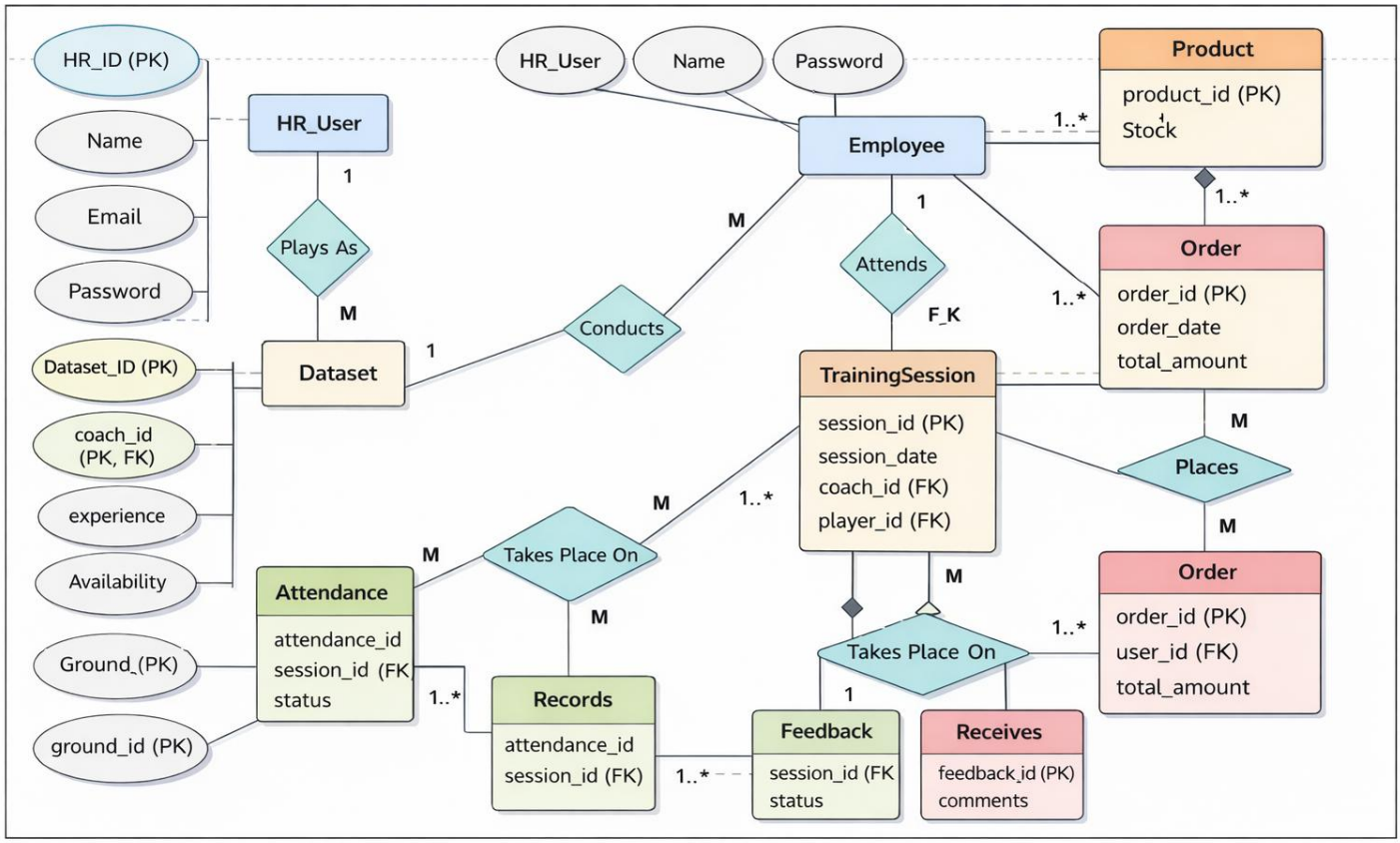
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M
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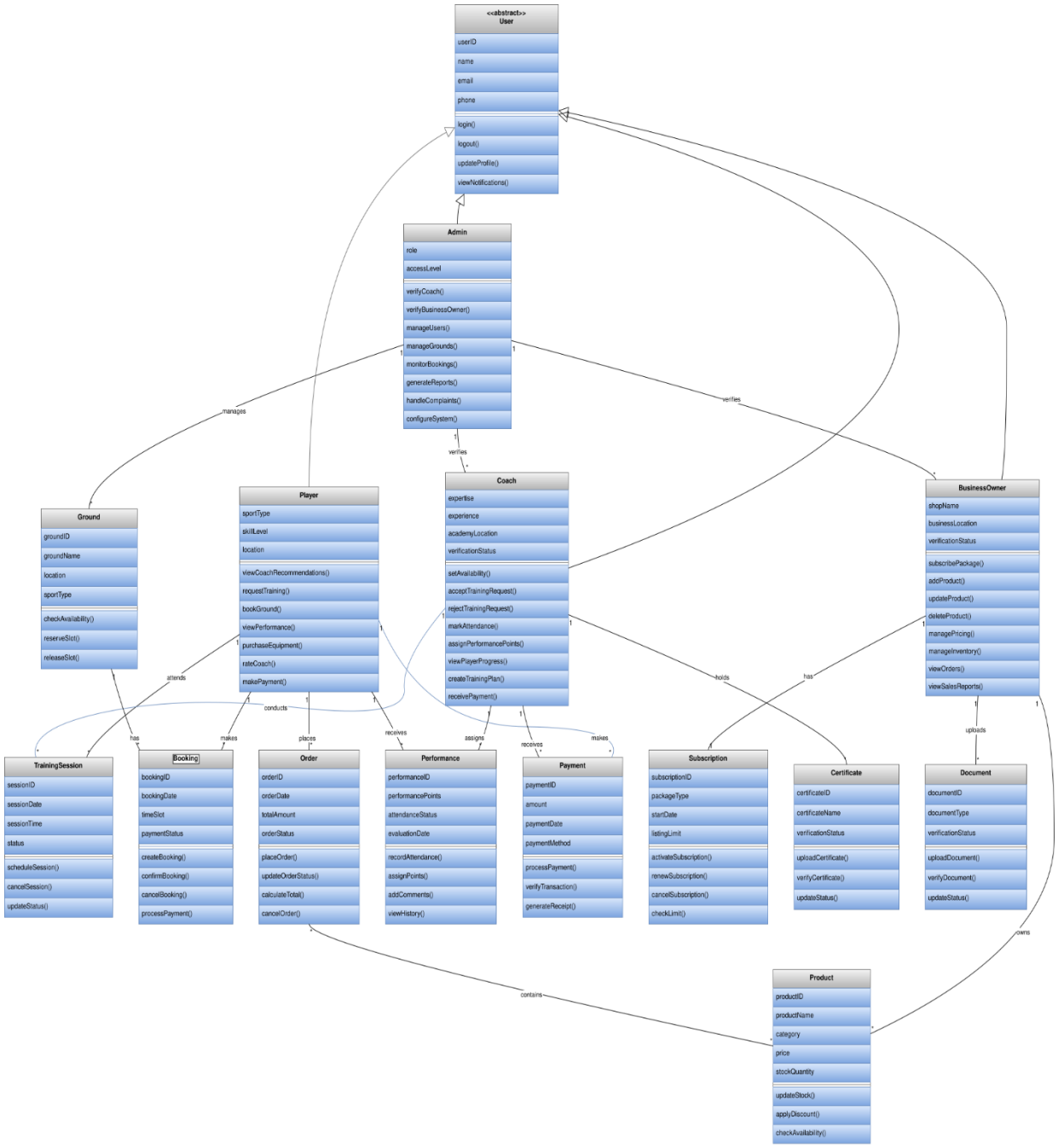
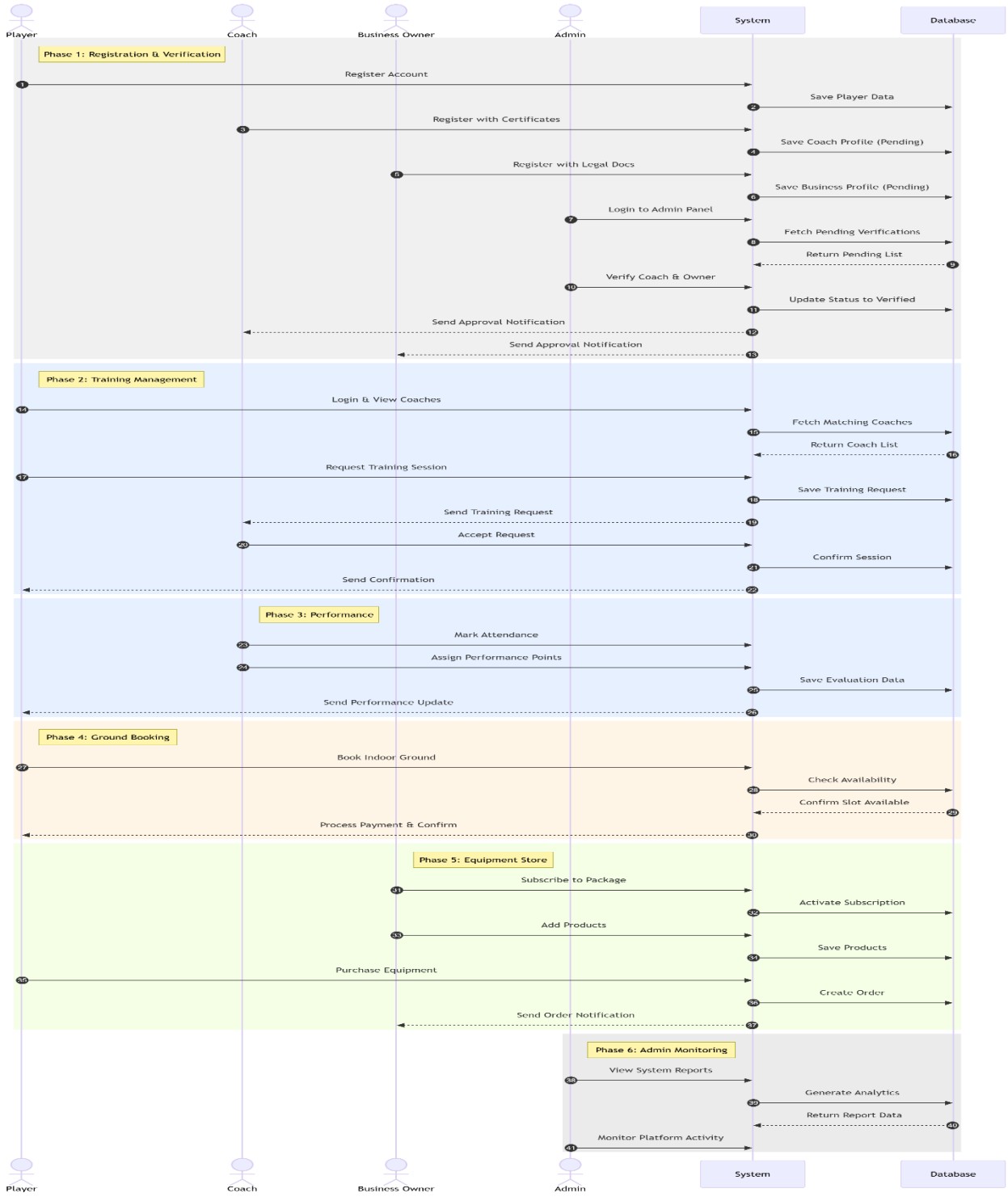
ABSTRACT CLASS Diagram



SEQUENCE DIAGRAM



COMPREHENSIVE USE CASES

Coach and Business Owner Use Cases

COACH USE CSES

1.1 UC-C1: Coach Registration and Login

Identifier: UC-C1

Purpose: To allow coaches to register with verification documents and login securely

Priority: High

Pre-conditions:

• Coach has access to the internet

• Coach has valid certification documents and academy location details

Post-conditions:

• Coach account is created and pending admin verification

Typical Course of Action

S#	Actor Action	System Response
1	Coach select account as coach choice	System shows coach sign_up form
2	Coach provide profile data (name, email, expertise, experience, sport type, academy location)	System validates entered information
3	Coach uploads verification certificates and location proof (Google Maps authentication or similar)	System accepts and stores documents for admin review
4	Coach submits registration	System creates account with pending status and notifies admin for verification
5	Coach receives notification	System sends email notification that account is under review

Alternate Course of Action

S#	Actor Action	System Response
1	Coach enters false or incomplete information	System displays validation errors and highlights required fields
2	Coach uploads invalid document format	System rejects upload and specifies acceptable formats (PDF, JPG, PNG)

1.2 UC-C2: Coach Profile Management

Identifier: UC-C2

Purpose: To allow coaches to manage their profile, expertise, experience, and verification documents

Priority: Medium

Pre-conditions:

• Coach successfully enters the system
• Coach account is verified by admin

Post-conditions:

• Coach profile is updated successfully

Typical Course of Action

S#	Actor Action	System Response
1	Coach navigates to profile management	System shows existing coach profile with all details
2	Coach updates expertise, experience, sport type, or location	System validates updated information
3	Coach uploads new certificates if needed	System stores documents and marks them for admin re-verification if critical changes
4	Coach saves changes	System updates profile and displays confirmation

1.3 UC-C3: Set Availability and Training Schedule

Identifier: UC-C3

Purpose: To allow coaches to define their availability for training sessions

Priority: High

Pre-conditions:

• Coach successfully enters the system

• Coach account is verified

Post-conditions:

• Coach availability is updated and visible for player recommendations

Typical Course of Action

S#	Actor Action	System Response
1	Coach access to availability management	System shows calendar interface with current availability
2	Coach selects days and time slots for availability	System highlights selected time slots
3	Coach sets recurring schedule or one-time availability	System saves schedule pattern
4	Coach confirms availability schedule	System saves availability and makes coach visible for player recommendations during those times

1.4 UC-C4: Receive Player Recommendations from System

Identifier: UC-C4

Purpose: To allow coaches to view players recommended by the system based on matching criteria

Priority: Medium

Pre-conditions:

• Coach successfully enters the system

• Coach has set availability and profile is complete

Post-conditions:

• Coach views list of recommended players

Typical Course of Action

S#	Actor Action	System Response
1	Coach enters into the section of player recommendations.	System shows list of players according to sport type, location and availability of coach.
2	Coach views player profiles	System shows player skill level, training goals, and location
3	Coach can send training invitations to recommended players	System facilitates outreach to potential students

1.5 UC-C5: Manage Training Session Requests

Identifier: UC-C5

Purpose: To allow coaches to view, accept, or reject training session requests from players

Priority: High

Pre-conditions:

• Coach is logged

• Players have sent training requests

Post-conditions:

• Training request is approved or rejected and player is notified

Typical Course of Action

S#	Actor Action	System Response
1	Coach views pending training requests	System displays list of player requests with player details and requested time slots
2	Coach reviews player profile and request details	System shows complete player information including skill level and training goals
3	Coach accepts or rejects request with optional message	System updates request status
4	Coach confirms schedule	System notifies the player of decision and creates training session if approved

Alternate Course of Action

S#	Actor Action	System Response
1	Coach's schedule has conflict with request	System highlights conflict and suggests alternative times

2	Coach is at maximum student capacity	System warns coach about capacity and allows override or rejection

1.6 UC-C6: Review and Approve Automated Training Plans

Identifier: UC-C6

Purpose: To allow coaches to review system-generated weekly training plans and make modifications

Priority: Medium

Pre-conditions:

• Coach successfully enters the system

• Training sessions have been scheduled

Post-conditions:

• Training plan is reviewed and approved or modified

Typical Course of Action

S#	Actor Action	System Response
1	Coach goes to training plans section.	System shows automatic weekly training programs of every player.
2	Coach reviews suggested exercises, duration, and goals	System shows plan details with rationale
3	Coach approves plan or makes modifications	System saves changes and marks plan as coach-reviewed
4	Coach finalizes training plan	System makes plan available to player and schedules notifications

1.7 UC-C7: Mark Player Attendance

Identifier: UC-C7
Purpose: To allow coaches to record attendance for training sessions
Priority: High
Pre-conditions:
• Coach successfully enters the system
• Training session has been conducted or is in progress

Post-conditions:
• Attendance record is saved in the system

Typical Course of Action

S#	Actor Action	System Response
1	Coach moves to attendance management	System shows list of scheduled sessions and enrolled players
2	Coach selects session and marks each player as present or absent	System records attendance status for each player
3	Coach adds notes for absent players if needed	System stores attendance notes
4	Coach saves attendance	System stores attendance records and updates player attendance history

1.8 UC-C8: Assign Weekly Performance Points

Identifier: UC-C8

Purpose: To allow coaches to evaluate players and assign weekly performance points

Priority: High

Pre-conditions:

• Coach successfully enters the system

• Training sessions have been conducted

Post-conditions:

• Performance points and evaluations are recorded and visible to player

Typical Course of Action

S#	Actor Action	System Response
1	Coach moves to performance evaluation section	System shows list of players and their current performance metrics
2	Coach selects a player to evaluate	System shows player's recent attendance, effort, and skill progression
3	Coach assigns performance points for various aspects (technique, fitness, attitude, etc.)	System validates point ranges and calculates total score
4	Coach adds evaluation comments and identifies weak points	System saves detailed evaluation
5	Coach submits evaluation	System records performance data, updates player dashboard, and notifies player

1.9 UC-C9: View Player Progress and Performance History

Identifier: UC-C9
Purpose: To allow coaches to track individual player development over time
Priority: Medium
Pre-conditions:
• Coach successfully enters the system
• Players have completed multiple training sessions

Post-conditions:
• Coach views comprehensive player progress data

Typical Course of Action

S#	Actor Action	System Response
1	Coach navigates to player progress section	System displays list of all assigned players
2	Coach selects a player to review	System shows historical performance graphs, attendance trends, and skill development
3	Coach analyzes progress patterns and areas needing attention	System provides analytics and comparison with player's goals
4	Coach uses insights to plan future training	System offers to adjust training plan based on progress data

1.10 UC-C10: Manage Booked Indoor Grounds

Identifier: UC-C10

Purpose: To allow coaches to view and manage ground bookings for their training sessions

Priority: Medium

Pre-conditions:

• Coach successfully enters the system

• Coach has training sessions scheduled

Post-conditions:

• Coach views ground booking information

Typical Course of Action

S#	Actor Action	System Response
1	Coach gets forward to bookings section.	System shows all the grounds that were booked to coach.
2	Coach views booking details (date, time, location, sport type)	System shows complete booking information and directions
3	Coach checks for booking conflicts or issues	System highlights any scheduling conflicts or cancellations

1.11 UC-C11: View Ratings and Feedback from Players

Identifier: UC-C11

Purpose: To allow coaches to see player ratings and feedback on their coaching quality

Priority: Medium

Pre-conditions:

• Coach successfully enters the system

• Players have provided ratings and feedback

Post-conditions:

• Coach views all ratings and feedback

Typical Course of Action

S#	Actor Action	System Response
1	Coach get forward to ratings and feedback section	System displays overall rating, rating distribution, and individual feedback
2	Coach reads detailed feedback from players	System shows feedback with player names (if not anonymous) and dates
3	Coach analyzes feedback patterns	System provides summary of common themes and areas of strength/improvement
4	Coach can respond to feedback	System records coach responses and notifies players

1.12 UC-C12: Receive Payment from Players

Identifier: UC-C12

Purpose: To allow coaches to receive training fees from players via the platform

Priority: High

Pre-conditions:

• Coach successfully enters the system

• Coach account has payment details configured

• Players have made payments

Post-conditions:

• Coach views payment transactions and account balance

Typical Course of Action

S#	Actor Action	System Response
1	Coach get forward to payments section	System shows payment dashboard including current balance and history of payment transactions
2	Coach views individual payment transactions	System shows payment details (player name, amount, date, status)
3	Coach requests withdrawal to bank account	System processes withdrawal request and provides confirmation
4	Coach receives payment confirmation	System sends notification and updates balance after successful transfer

1.13 UC-C13: Create and Modify Weekly Training Plans

Identifier: UC-C13

Purpose: To allow coaches to create customized weekly training plans for individual players

Priority: High

Pre-conditions:

• Coach successfully enters the system

• Coach has accepted player for training

Post-conditions:

• Weekly training plan is created and assigned to player

Typical Course of Action

S#	Actor Action	System Response
1	Coach get forward to training plan creation	System displays plan creation interface with templates
2	Coach selects player and training week	System shows player's current skill level and goals
3	Coach adds exercises, drills, duration, and objectives	System validates plan structure
4	Coach sets performance targets for the week	System records targets for evaluation
5	Coach saves and assigns training	System makes plan available to player and sends msgs

1.14 UC-C14: Receive and Manage Notifications

Identifier: UC-C14

Purpose: To allow coaches to receive system notifications about requests, approvals, and updates

Priority: Medium

Pre-conditions:

• Coach successfully enters the system

Post-conditions:

• Coach receives and views notifications

Typical Course of Action

S#	Actor Action	System Response
1	System generates notification for coach	System creates notification for relevant events (training request, payment received, admin verification, player feedback)
2	Coach receives notification	System displays notification in coach dashboard and sends email/SMS
3	Coach views notification details	System shows complete notification with action links if applicable
4	Coach acts on notification (optional)	System processes coach action and updates relevant records

1.15 UC-C15: Upload and Update Certification Documents

Identifier: UC-C15
Purpose: To allow coaches to add or update their professional certifications
Priority: Medium
Pre-conditions:
• Coach successfully enters the system

Post-conditions:
• New or updated certificates are uploaded and pending admin verification

Typical Course of Action

S#	Actor Action	System Response
1	Coach move forward to certification management	System shows current certificates and their verification status
2	Coach uploads new certificate or updates existing one	System accepts document upload and validates file format
3	Coach adds certificate details (name, issuing authority, date, expiry)	System stores certificate metadata
4	Coach submits for verification	System marks certificate as pending verification and notifies admin
5	Coach receives verification result	System notifies coach when admin approves or rejects certificate

2. BUSINESS OWNER USE CASES

2.1 UC-B1: Business Owner Registration and Login

Identifier: UC-B1

Purpose: To allow business owners to register with legal documents and login securely

Priority: High

Pre-conditions:

• There is internet access by the business owner.

• Business owner has valid legal documents and shop details

Post-conditions:

• Business owner account is created and pending admin verification

Typical Course of Action

S#	Actor Action	System Response
1	Business owner selects register as business owner option	System displays business registration form
2	Business owner enters business profile (name, email, shop name, location, business type)	System validates entered information
3	Business owner uploads legal documents (business license, tax registration, etc.)	System accepts and stores documents for admin review
4	Business owner submits registration	System creates account with pending status and notifies admin for verification
5	Business owner receives notification	System sends email notification that account is under review

Alternate Course of Action

S#	Actor Action	System Response
1	Business owner enters invalid business information	System displays validation errors and highlights issues
2	Business owner uploads incomplete documentation	System lists required documents that are missing

2.2 UC-B2: Business Profile Management

Identifier: UC-B2

Purpose: To allow business owners to manage their business profile and shop details

Priority: Medium

Pre-conditions:

• The owner of the business is logged into the system.

• Business account is verified by admin

Post-conditions:

• Profile of business is modified.

Typical Course of Action

S#	Actor Action	System Response

1	Business owner movess to profile management	System shows current business profile with all details
2	Business owner updates shop details, location, or contact information	System validates updated information
3	Business owner uploads updated legal documents if needed	System stores documents and marks them for admin re-verification if critical changes
4	Business owner saves changes	System updates profile and displays confirmation

2.3 UC-B3: Subscribe to Product Listing Package

Identifier: UC-B3

Purpose: To allow business owners to subscribe to listing packages (Basic, Pro, Premium) for showcasing products

Priority: High

Pre-conditions:

• Business owner has a registered and verified account

• Business owner has valid payment method

Post-conditions:

• Subscription is activated and listing limit is updated based on selected package

Typical Course of Action

S#	Actor Action	System Response
1	Business owner is subscribing (or expired) to the business)	System displays package details with pricing and features
2	Business owner reviews package details and pricing	System shows monthly cost and benefits comparison
3	Business owner enters payment card details	System verifies payment information securely
4	Business owner confirms subscription	System processes payment and activates subscription
5	Business owner receives confirmation	System sends confirmation email with invoice, updates listing limits, and enables product management

Alternate Course of Action

S#	Actor Action	System Response
1	Payment fails at the time of processing	System displays error message and allows retry with different payment method
2	Business owner already has active subscription	System offers upgrade/downgrade options instead of new subscription

2.4 UC-B4: Manage and Renew Subscription

Identifier: UC-B4

Purpose: To allow business owners to view, upgrade, downgrade, or cancel their subscriptions

Priority: High

Pre-conditions:

• Business owner is logged into the system

• Business owner has an active or expired subscription

Post-conditions:

• Subscription status is updated according to business owner action

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to subscription management	System displays current subscription details (package, expiry date, listing usage)
2	Business owner selects action (renew, upgrade, downgrade, cancel)	System shows relevant options and pricing
3	Business owner confirms action and processes payment if needed	System updates subscription, processes payment, and adjusts listing limits
4	Business owner receives confirmation	System sends email confirmation and updates account immediately

Alternate Course of Action

S#	Actor Action	System Response
1	Business owner attempts to downgrade but has more products than new limit	System warns about exceeding limit and prompts to remove excess products first

2.5 UC-B5: Create and Configure Online Equipment Store

Identifier: UC-B5
Purpose: To allow business owners to set up their online sports equipment store
Priority: High
Pre-conditions:
• Business owner is enetr nto the system
• Business owner has active subscription

Post-conditions:
• Online store is created and configured

Typical Course of Action

S#	Actor Action	System Response
1	In order to permit the owners of the business to put on new sports equipments in their store.	System displays store configuration wizard
2	Business owner enters store details (name, description, policies, shipping info)	System validates store information
3	Business owner uploads store logo and banner images	System stores branding assets
4	Business owner configures payment and shipping settings	System saves configuration
5	Business owner activates store	System makes store visible to customers and displays store URL

2.6 UC-B6: Add Sports Products (Create)

Identifier: UC-B6

Purpose: To allow business owners to add new sports equipment products to their store

Priority: High

Pre-conditions:

• Business owner is logged into the system

• Business owner has active subscription with available listing slots

Post-conditions:

• Product is added to store and visible to customers

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to product management	System displays product dashboard and available listing count
2	Clicks on business owner add product	System displays product creation form
3	Business proprietor keeps product information (name, category, description, price, stock quantity)	Product creation form is displayed on system
4	Business owner uploads product images	System stores images and generates thumbnails
5	Business owner saves product	System creates product listing, decrements available slots, and makes product visible in store

Alternate Course of Action

S#	Actor Action	System Response
1	Business owner has reached listing limit	System prevents adding new product and prompts to upgrade subscription

2.7 UC-B7: Update Sports Products

Identifier: UC-B7

Purpose: To allow business owners to modify existing product information

Priority: Medium

Pre-conditions:

• Business owner is logged into the system

• Products exist in the store

Post-conditions:

• Updating of product information is accomplished.

Typical Course of Action

S#	Actor Action	System Response
1	Owner of business visits the product management.	System displays list of all products
2	Business owner chooses a product to be edited.	System displays product details in editable form
3	Business owner modifies product information (price, stock, description, images)	System validates changes
4	Business owner saves changes	System updates product, maintains version history, and reflects changes immediately in store

2.8 UC-B8: Delete Sports Products

Identifier: UC-B8

Purpose: To enable business owners to take off products in the store.

Priority: Medium

Pre-conditions:

• Business owner is logged into the system

• Products exist in the store

Post-conditions:

• Product is removed from store and listing slot is freed

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to product management	System displays list of all products
2	Business owner selects product to delete	System displays product details and delete confirmation
3	Business owner confirms deletion	System checks for pending orders with this product
4	Business owner proceeds with deletion	System removes product, frees listing slot, and archives product data

Alternate Course of Action

S#	Actor Action	System Response
1	Product has pending orders	System warns business owner and suggests marking as out of stock instead of deleting

2.9 UC-B9: Manage Product Pricing

Identifier: UC-B9

Purpose: To allow business owners to set and update product prices and discounts

Priority: Medium

Pre-conditions:

• Business owner is logged into the system

• Products exist in the store

Post-conditions:

• Product pricing is updated successfully

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to pricing management	System displays products with current pricing
2	Business owner updates prices or sets discounts	System validates pricing (no negative values, discount doesn't exceed price)
3	Business owner sets sale period for discounts (optional)	System schedules automatic price changes
4	Business owner saves pricing changes	System updates prices and displays new values in store immediately

2.10 UC-B10: Manage Product Stock and Inventory

Identifier: UC-B10

Purpose: To allow business owners to track and update product inventory levels

Priority: High

Pre-conditions:

• Business owner is logged into the system

• Products exist in the store

Post-conditions:

• Stock levels are updated and reflected in store availability

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to inventory management	System displays all products with current stock levels and low stock alerts
2	Business owner updates stock quantities	System validates stock values (non-negative)
3	Business owner sets low stock threshold for alerts	System configures automatic notifications
4	Business owner saves stock changes	System updates inventory, adjusts product availability, and sends low stock alerts if needed

2.11 UC-B11: Manage Product Images

Identifier: UC-B11

Purpose: To allow business owners to upload, update, and organize product images

Priority: Medium

Pre-conditions:

• Business owner is logged into the system

• Products exist in the store

Post-conditions:

• Product images are updated successfully

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to product image management	System displays products and their current images
2	Business owner selects product to update images	System shows image gallery for that product
3	Business owner uploads new images or reorders existing ones	System validates image formats and sizes
4	Business owner sets primary image	System saves image changes and regenerates thumbnails

2.12 UC-B12: View and Manage Customer Orders

Identifier: UC-B12

Purpose: To allow business owners to view orders placed by customers

Priority: High

Pre-conditions:

• Business owner is logged into the system

• Customers have placed orders

Post-conditions:

• Business owner views order details and can update order status

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to orders section	System displays list of all orders with status filters (pending, processing, shipped, completed)
2	Business owner selects an order to view details	System shows complete order information (customer, products, quantities, prices, shipping address)
3	The update of images of products is successful.	Status of orders updated by the business owner.
4	Business owner adds tracking information for shipped orders	System saves order updates and notifies customer

2.13 UC-B13: View Basic Sales Reports

Identifier: UC-B13

Purpose: To allow business owners to view sales analytics and performance metrics

Priority: Medium

Pre-conditions:

• Business owner is logged into the system

• Sales transactions have occurred

Post-conditions:

• Business owner views sales reports and analytics

Typical Course of Action

S#	Actor Action	System Response
1	Business owner moves to reports section	System displays dashboard with key sales metrics (total revenue, orders, popular products)
2	The options of report type and date range are selected by the business owner.	System produces report of sales information, tendencies and comparisons.
3	Business owner analyzes sales patterns	System provides visual charts and graphs
4	Business owner exports report (optional)	System generates downloadable report in PDF or Excel format

2.14 UC-B14: Hire or Connect with Coaches

Identifier: UC-B14

Purpose: To allow business owners to establish partnerships or sponsorships with coaches

Priority: Low

Pre-conditions:

• Business owner enter into the system

• Coaches are registered in the system

Post-conditions:

• Connection request is sent to coach

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to coach directory	System displays list of verified coaches with their specialties and locations
2	Business owner searches for coaches by sport type or location	System filters and displays matching coaches
3	Business owner views coach profile	System shows coach details, ratings, and player count
4	Business owner sends partnership or sponsorship	System delivers message to coach and awaits response

2.15 UC-B15: Receive Approval and Reminder Notifications

Identifier: UC-B15

Purpose: To allow business owners to receive notifications about verification, orders, and subscription status

Priority: Medium

Pre-conditions:

• Business owner is registered in the system

Post-conditions:

• Business owner receives and views notifications

Typical Course of Action

S#	Actor Action	System Response
1	System generates notification for business owner	System creates notification for events (admin approval/rejection, new order, subscription renewal reminder, low stock alert)
2	Business owner receives notification	System displays notification in dashboard and sends email
3	Business owner views notification details	System shows complete notification with action links
4	Business owner acts on notification if needed	System processes action (e.g., view order, renew subscription)

2.16 UC-B16: Upload and Update Verification Documents

Identifier: UC-B16

Purpose: To enable the owners of businesses to add or edit their legal and business documents

Priority: Medium

Pre-conditions:

• The system logs the business owner into it.

Post-conditions:

• Documents are uploaded and pending admin verification

Typical Course of Action

S#	Actor Action	System Response
1	Business owner navigates to document management	System displays current documents and their verification status
2	Business owner uploads new document or updates existing one	System accepts document upload and validates file format
3	Business owner adds document details (type, issue date, expiry date)	System stores document metadata
4	Business owner submits for verification	System marks documents as pending verification and notifies admin
5	Business owner receives results	System notifies owner when admin approves or rejects

COMPREHENSIVE USE CASES

Player and Admin Use Cases

3. PLAYER USE CASES

3.1 UC-P1: Player Registration and Login

Identifier: UC-P1

Purpose: To allow a player to register and securely log into the system

Priority: High

Pre-conditions:

• The internet is accessible to the user.

• The player is not already registered (for registration)

Post-conditions:

• The player account is successfully created or logged in

Typical Course of Action

S#	Actor Action	System Response
1	Player chooses register or log in option.	Registration/login form is displayed in the system.
2	Player enters required information (name, email, password, sport selection, skill level, location)	System validates entered data
3	Player submits the form	System creates account or logs in the player
4	Player accesses dashboard	System displays player dashboard with profile summary

Alternate Course of Action

S#	Actor Action	System Response
1	Player keying in unauthorized or valid email.	System provides relevant error message.
2	Player retries login or registration	System redisplays login/registration form

3.2 UC-P2: Player Profile Management

Identifier: UC-P2

Purpose: To allow players to update their profile information and preferences

Priority: Medium

Pre-conditions:

• Player is logged into the system

Post-conditions:

• Player profile is updated successfully

Typical Course of Action

S#	Actor Action	System Response
1	Player goes to the profile management section.	Profile displays prevailing profile details.
2	Player updates personal details, sport selection, skill level, or location	System validates the updated information
3	Player saves changes	System updates profile and displays confirmation message

3.3 UC-P3: Automated Coach Recommendation

Identifier: UC-P3

Purpose: To recommend suitable coaches based on player profile, sport type, skill level, and location

Priority: High

Pre-conditions:

• Player is entered the system

• Player profile is completed with sport type and location

Post-conditions:

• A list of recommended coaches is displayed to the player

Typical Course of Action

S#	Actor Action	System Response
1	Player requests coach recommendations	System analyzes player profile, sport type, skill level, and location
2	Player views the list of recommended coaches	System filters and displays matching verified coaches with their expertise, experience, and ratings
3	Player views detailed coach profiles	System displays complete coach information including certifications, availability, and player feedback

Alternate Course of Action

S#	Actor Action	System Response
1	No coach is available for the selected sport and location	System displays availability notification and suggests nearby alternatives

3.4 UC-P4: Request Training Session

Identifier: UC-P4

Purpose: To allow a player to request a training session with a selected coach

Priority: High

Pre-conditions:

• Player is logged in

• A coach has been selected from recommendations

Post-conditions:

• Training request is sent to the selected coach

Typical Course of Action

S#	Actor Action	System Response
1	Player chooses a coach and clicks request training session.	System displays coach availability in request form.
2	Player selects preferred date and time	System checks coach availability for selected slot
3	Player submits request with optional message	System sends request to coach and displays confirmation
4	Player receives notification	System notifies player of request status (pending, accepted, rejected)

Alternate Course of Action

S#	Actor Action	System Response
1	Selected time slot is unavailable	System suggests alternative available time slots

3.5 UC-P5: View Training Schedule and Weekly Plans

Identifier: UC-P5

Purpose: To allow players to view their training schedules and automated weekly training plans

Priority: Medium

Pre-conditions:

• Player is logged into the system

• At least one training session has been approved

Post-conditions:

• Player views their complete training schedule and weekly plans

Typical Course of Action

S#	Actor Action	System Response
1	Player navigates to training schedule section	System displays calendar view of upcoming training sessions
2	Player selects a specific session or week	System displays detailed training plan including exercises, duration, and coach notes
3	Player views assigned coach and location details	System shows complete session information with coach contact and venue

3.6 UC-P6: Indoor Ground Booking

Identifier: UC-P6

Purpose: To allow players to book indoor sports grounds for cricket and badminton

Priority: High

Pre-conditions:

• Player is logged in

• There are ground and time slot.

Post-conditions:

• Booking confirmation is generated and ground is reserved

Typical Course of Action

S#	Actor Action	System Response
1	Player selects ground booking option	System displays available grounds filtered by location and sport type
2	Player selects sport type (cricket/badminton), date, and time	System checks real-time slot availability and pricing
3	Player makes reservation and goes to checkout.	System processes payment and reserves the ground
4	Player receives booking confirmation	System saves booking, sends confirmation email, and displays booking details

Alternate Course of Action

S#	Actor Action	System Response
1	Chosen slot has already been booked.	System displays error message and shows alternative available slots
2	Payment fails	System releases the temporary reservation and prompts player to retry payment

3.7 UC-P7: View Performance Points and Progress

Identifier: UC-P7

Purpose: To allow players to track their performance points, attendance, and overall progress

Priority: Medium

Pre-conditions:

• Player is logged into the system

• Training sessions have been conducted

Post-conditions:

• Player views their performance metrics and progress history

Typical Course of Action

S#	Actor Action	System Response
1	Player navigates to dashboard or performance section	System displays performance dashboard with graphs and statistics
2	Player views weekly performance points	System shows points assigned by coach with detailed breakdown
3	Player checks attendance record and weak points	System displays attendance history, identified weak areas, and improvement suggestions
4	Player views progress over time	System presents visual charts showing performance trends

3.8 UC-P8: Purchase Sports Equipment

Identifier: UC-P8

Purpose: To allow players to browse and purchase sports equipment from business owner stores

Priority: Medium

Pre-conditions:

• Player is logged into the system

• Products are available in the marketplace

Post-conditions:

• Order is placed and payment is processed

Typical Course of Action

S#	Actor Action	System Response
1	Player navigates to equipment store	System displays product listings with filters and categories
2	Player searches and selects products	System shows product details, images, pricing, and availability
3	Player adds items to cart and proceeds to checkout	System displays cart summary and checkout form
4	Player completes payment	System processes payment, creates order, and sends confirmation

3.9 UC-P9: Rate and Provide Feedback to Coach

Identifier: UC-P9

Purpose: To allow players to rate coaches and provide feedback on training quality

Priority: Medium

Pre-conditions:

• Player is logged into the system

• At least one training session has been completed

Post-conditions:

• Feedback is recorded and visible to coach and admin

Typical Course of Action

S#	Actor Action	System Response
1	Player navigates to coach profile or completed sessions	System displays feedback/rating option
2	Player selects rating (1-5 stars) and writes feedback	System validates input
3	Player submits feedback	System saves feedback, updates coach rating, and notifies coach

3.10 UC-P10: Player Payment to Coach

Identifier: UC-P10

Purpose: To allow players to pay training fees to coaches via secure card payments

Priority: High

Pre-conditions:

• Player has a registered account

• Player has selected a coach for training

Post-conditions:

• Payment is processed, coach account is credited, and transaction record is generated

Typical Course of Action

S#	Actor Action	System Response
1	Player selects coach and clicks 'Pay Fee'	System displays secure payment gateway with fee amount
2	Player enters card details and amount	System verifies card details and validates amount
3	Player confirms payment	System processes payment and updates coach account balance
4	System sends confirmation	System records transaction, sends receipt to player and notification to coach

3.11 UC-P11: Receive System Notifications

Identifier: UC-P11

Purpose: To notify players about coach assignments, booking status, and training reminders

Priority: Medium

Pre-conditions:

• Player is registered in the system

Post-conditions:

• Player receives and views notifications

Typical Course of Action

S#	Actor Action	System Response
1	System triggers notification event	System generates notification for relevant event (booking confirmation, coach assignment, session reminder)
2	Player receives notification	System displays notification in player dashboard and sends email/SMS
3	Player views notification details	System shows complete notification with action links if applicable

4. ADMIN USE CASES

4.1 UC-A1: Admin Login

Identifier: UC-A1

Purpose: To provide secure authentication for system administrators

Priority: High

Pre-conditions:

• Admin has valid credentials

• Admin panel is accessible

Post-conditions:

• Admin is successfully logged in and dashboard is displayed

Typical Course of Action

S#	Actor Action	System Response
1	Admin has been logged successfully and dashboard has been displayed.	System shows login form of the administrator.
2	Username and password inputs are made (admin).	System authenticates authorization.
3	Admin submits login form	System authenticates admin and grants access to admin panel
4	Admin accesses dashboard	System displays admin dashboard with system statistics

Alternate Course of Action

S#	Actor Action	System Response
1	Invalid credentials are typed in by Admin.	System will display the error message and to restart.

4.2 UC-A2: View Admin Dashboard and Statistics

Identifier: UC-A2

Purpose: To give administrators a view of activity and most indicative metrics of the system.
Priority: High

Pre-conditions:

• Admin is logged into the system

Post-conditions:

• Admin views comprehensive system statistics and activity

Typical Course of Action

S#	Actor Action	System Response

1	Admin accesses dashboard	System displays key metrics (total users, coaches, business owners, bookings, revenue)
2	Admin views recent activities	System shows recent registrations, verifications, bookings, and transactions
3	Admin checks pending verifications	System displays count and list of pending coach and business owner verifications
4	Admin reviews system health indicators	System shows server status, database performance, and error logs

4.3 UC-A3: Verify and Approve Coach

Identifier: UC-A3

Purpose: To allow admin to verify coach credentials, certificates, and academy location before granting access

Priority: High

Pre-conditions:

• Admin is logged into the system

• Coach has submitted registration with required documents

Post-conditions:

• Coach verification status is updated and coach is notified

Typical Course of Action

S#	Actor Action	System Response
1	Admin moves to coach verification section	System shows list of pending coach verifications
2	Admin selects a coach to review	System displays coach profile, uploaded certificates, and location details
3	Admin verifies certificates and checks academy location (via Google Maps or similar)	System shows document viewer and location verification tools
4	Admin approves or rejects coach with comments	System updates coach status to verified/rejected
5	System sends notification to coach	System sends email notification about verification decision

Alternate Course of Action

S#	Actor Action	System Response
1	Admin requests additional documents from coach	System marks verification as pending and notifies coach to upload more documents

4.4 UC-A4: Verify and Approve Business Owner

Identifier: UC-A4

Purpose: To allow admin to verify business owner credentials and legal documents before granting marketplace access

Priority: High

Pre-conditions:

• Admin is logged into the system

• Business owner has submitted registration with legal documents

Post-conditions:

• Business owner verification status is updated and owner is notified

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to business owner verification section	System displays list of pending business owner verifications
2	Admin selects a business owner to review	System displays business profile, uploaded legal documents, and shop details
3	Admin verifies legal documents and business legitimacy	System provides document viewer and verification checklist
4	Admin approves or rejects business owner with comments	System updates business owner status to verified/rejected
5	System sends notification to business owner	System sends email about verification decision and next steps

Alternate Course of Action

S#	Actor Action	System Response
1	Admin requests additional documentation	System marks verification as pending and notifies owner to provide more documents

4.5 UC-A5: Manage Users (CRUD Operations)

Identifier: UC-A5

Purpose: To allow admin to create, read, update, and delete user accounts

Priority: High

Pre-conditions:

• Admin is logged into the system

Post-conditions:

• User data is successfully managed (created, updated, or deleted)

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to user management section	System displays list of all registered users with search and filter options
2	Admin selects operation (create, view, update, or delete)	System displays appropriate form or user details
3	Admin performs desired operation	System validates input and executes operation
4	Admin confirms changes	System updates database and displays confirmation message

4.6 UC-A6: Manage Coaches (CRUD Operations)

Identifier: UC-A6

Purpose: To allow admin to manage coach accounts and profiles

Priority: High

Pre-conditions:

• Admin is logged into the system

Post-conditions:

• Coach data is successfully managed

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to coach management section	System displays list of all coaches (verified, pending, rejected)
2	Admin selects a coach to view or edit	System displays complete coach profile with verification status and documents
3	Admin updates coach information or status	System validates changes
4	Admin saves changes or deletes coach account	System updates database and notifies coach if status changed

4.7 UC-A7: Manage Business Owners (CRUD Operations)

Identifier: UC-A7

Purpose: To allow admin to manage business owner accounts and store permissions

Priority: High

Pre-conditions:

• Admin is logged into the system

Post-conditions:

• Business owner data is successfully managed

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to business owner management section	System displays list of all business owners with their subscription status
2	Admin selects a business owner to review	System displays business profile, products, and sales data
3	Admin updates business information, subscription, or listing limits	System validates changes
4	Admin saves changes or reject	System updates database and notifies business owner

4.8 UC-A8: Manage Sports Categories

Identifier: UC-A8

Purpose: To allow admin to manage supported sports categories (cricket and badminton)

Priority: Medium

Pre-conditions:

• Admin is logged into the system

Post-conditions:

• Sports category settings are updated

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to sports category management	System displays current sports categories (cricket, badminton)
2	Admin views or updates category settings	System shows category configuration options
3	Admin modifies category parameters	System validates settings
4	Admin saves changes	System updates category configuration and applies changes system-wide

4.9 UC-A9: Manage Indoor Grounds and Time Slots

Identifier: UC-A9

Purpose: To allow admin to add, update, or remove indoor sports grounds and their available time slots

Priority: High

Pre-conditions:

• Admin is logged into the system

Post-conditions:

• Ground information and time slots are successfully managed

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to ground management section	System displays list of all registered grounds with their details
2	Admin adds new ground or selects existing ground	System displays ground form with fields for name, location, sport type, capacity, and pricing
3	Admin configures time slots and availability	System provides time slot management interface
4	Admin saves ground information	System validates data, updates database, and makes ground available for booking

4.10 UC-A10: Monitor Bookings and Training Sessions

Identifier: UC-A10

Purpose: To allow admin to view and monitor all ground bookings and training sessions

Priority: Medium

Pre-conditions:

• Admin is logged into the system

Post-conditions:

• Admin views comprehensive booking and session data

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to bookings monitoring section	System displays calendar view of all bookings and training sessions
2	Admin filters by date, ground, sport type, or user	System applies filters and updates display
3	Admin views detailed booking information	System shows booking details including player, coach, ground, time, and payment status
4	Admin identifies conflicts or issues	System highlights any scheduling conflicts or payment issues

4.11 UC-A11: Monitor Performance Points and Attendance

Identifier: UC-A11

Purpose: To allow admin to track player performance evaluations and attendance records

Priority: Medium

Pre-conditions:

• Admin is logged into the system

• Training sessions have been conducted

Post-conditions:

• Admin views performance and attendance analytics

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to performance monitoring section	System displays overview of performance points and attendance across all players
2	Admin selects specific player or coach	System shows detailed performance history and attendance records
3	Admin reviews evaluation patterns	System displays analytics including average scores, improvement trends, and attendance rates
4	Admin identifies outliers or concerns	System highlights unusual patterns requiring attention

4.12 UC-A12: Handle Complaints and Dispute Management

Identifier: UC-A12

Purpose: To allow admin to receive, review, and resolve complaints and disputes between users

Priority: High

Pre-conditions:

• Admin is logged into the system

• Complaints have been submitted by users

Post-conditions:

• Complaint is reviewed and resolved with action taken

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to complaints section	System displays list of pending and resolved complaints
2	Admin selects a complaint to review	System shows complaint details, involved parties, and evidence
3	Admin investigates by reviewing user history and interactions	System provides access to relevant transaction and communication logs
4	Admin takes action (warning, suspension, refund, etc.)	System records decision and executes action
5	Admin notifies involved parties	System sends resolution notification to complainant and respondent

4.13 UC-A13: Generate Reports for System Monitoring

Identifier: UC-A13

Purpose: To allow admin to generate various reports for analysis and decision-making

Priority: Medium

Pre-conditions:

• Admin is logged into the system

• Sufficient data exists in the system

Post-conditions:

• Report is generated and available for download or viewing

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to reports section	System displays available report types (user statistics, revenue, bookings, performance, etc.)
2	Admin selects report type and parameters (date range, filters)	System shows report configuration options
3	Admin generates report	System processes data and creates report
4	Admin views or downloads report	System displays report in dashboard or provides download link (PDF, Excel)

4.14 UC-A14: Monitor Business Owner Subscription Payments

Identifier: UC-A14

Purpose: To allow admin to track and manage business owner subscription payments for listing packages

Priority: High

Pre-conditions:

• Admin is logged into the system

• Business owners have active subscriptions

Post-conditions:

• Admin views subscription status and payment history

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to subscription management	System displays all business owner subscriptions with status (active, expired, pending)
2	Admin views payment history for a business owner	System shows detailed transaction history including dates, amounts, and package types
3	Admin handles payment issues or disputes	System provides tools to adjust subscription status or issue refunds
4	Admin updates subscription manually if needed	System records manual changes and notifies business owner

4.15 UC-A15: Configure System Settings

Identifier: UC-A15
Purpose: To allow admin to configure global system settings and parameters
Priority: Medium
Pre-conditions:
• Admin is logged into the system with super admin privileges

Post-conditions:
• System settings are updated and applied

Typical Course of Action

S#	Actor Action	System Response
1	Admin navigates to system settings	System displays configuration options for various modules
2	Admin modifies settings (payment gateways, notification templates, pricing, etc.)	System validates settings
3	Admin saves configuration changes	System applies settings and logs the changes
4	System confirms update	System displays confirmation and restarts affected services

4. Non-Functional Requirements

The non-functional requirements of Sports Ecosystem Platform are presented in the next section.

These are requirements that specify the quality of the characteristics, limitations, and terms of operation of the system.

4.1 Performance Requirements

The system will have the following performance requirements to facilitate easy operation with everyone using it:

- With normal load, the application must have a capability to respond to your request (sign in, place a flight reservation) within 2-3 seconds.
- After request has been placed, the results of coach recommendation should be produced and displayed within 5 seconds.
- Ground Booking Availability Checks: In Ground booking, availability checks will be done in real time in order to avoid any duplicity.
- The system will be able to service more than one user at a time without any significant slowing of the performance.
- Read and write operations on the database will be streamlined so that they quickly retrieve the player profile, coach information and booking details.

These are performance requirements required to ensure a good user experience and to prevent delays when these are needed most like during training sessions requests and ground bookings which are time-sensitive operations.

4.2 Safety Requirements

Sports Ecosystem Platform is a programmable hardware and does not physically operate devices. Thus, safety requirements are actually limited in their main intentions on how to curb abuse and loss of data.

- The system will not allow backing up of essential information like user accounts, booking and training history by mistake.
- Confirmation prompts will be shown prior to making some unsalvageable moves like accounts deletion or booking cancellations.
- The system will have data backup to avert the loss in case the system fails.
- The unauthorized users will not be permitted to undertake actions which will impact on other users and system integrity.

These protections help in making sure that the system does not harm in an indirect way including loss of valuable training or booking information..

4.3 Security Requirements

The Sports Ecosystem Platform has a very high need of security owing to the sensitive nature of user data.

- The system will have authorization checks with secure user authentication of all user roles (Player, Coach, Business Owner, Admin).
- Role-based access control: This will be implemented, so that the user will be restricted to features of that role.
- User passwords will also be stored in an encrypted manner.
- The system will secure personal data of the user including contact data, and location data.
- Only authorized administrators should be allowed to perform administrative functions like verification, approval and generation of reports of this kind.
- The system will ensure that there is no unauthorized access to the database and backend services.
- These mandates guarantee confidentiality, integrity and controlled access of data in the system.

4.4 Additional Software Quality Attributes

The Sports Ecosystem Platform will meet the following software quality features:

Usability

- The user interface will be easy to use and non-technical users.
- All user panels will have the same type of navigation.

Reliability

- The application will not be affected with frequent failures when being used by ordinary people.
- The application will have the capability to recover without any data loss should it fail.

Maintainability

- The application would be modular in the sense that it is easy to carry out future programs together with fixes in the system.

Code structure:

- Code structure will be based on standard practice of programming.

Scalability

- The application can expand later i.e. by adding new sports categories or roles.

Portability

- The application will use the normal web browsers without necessarily installing software on the computer.

Availability

- The application can be access by user anytime unless the system is undergoing an envisaged maintenance period.

5. Other Requirements

The section contains the further requirements of the Sports Ecosystem Platform not explicitly mentioned in the previous sections of this Software Requirements Specification.

5.1 Database Requirements

- The application will entail the use of centralized database in which all the application data will be assembled in the form of player profile, coach profile, trainings sessions, ground bookings, attendance record, product listing and order.
- Each of the user roles (Player, Coach, Business Owner, Admin) will have its unique identifier in the database.
- The database will contain referential integrity among several entities like Player, Coach, Training session, booking, product and order.
- The system will ensure include CRUD operations actions on all key entities.
- The backup mechanisms will be implemented to avoid the loss of data.

5.2 External Interface Requirements

5.2.1 User Interface

- The system will have dedicated user interfaces (panels) with the Players, Coaches, Business Owners and Admins.
- The user interface can be access by standard web browsers
- Forms, dashboards and reports will be well labeled and user friendly.

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5.2.2 Software Interfaces

- The system will have connectivity with a backend server or to process requests and handle information.
- The next point is that the frontend will be connected directly to the backend with secure means of communication.
- The system will incorporate notification services like booking confirmation, requesting training and approval notifications.

5.2.3 Communication Interfaces

- The system will be using the ordinary internet protocols.
- Notifications will be presented in the system interface.

5.3 Hardware Requirements

- The system will operate on a server that is powerful enough in terms of processing power and memory to support a number of users.
- End users will demand an or an internet-enabled device with an updated web browser.
- No hardware is needed in order to operate the system.

5.4 Software Requirements

Various web development technologies will therefore be used in deploying the system :

- It shall be supported with the backend that will enable database connection and storm authorization.
- This system will be supported by the operating environments such as windows based and Linux based systems
- The database management platform will be of the relational type of data storage.

5.5 Legal and Regulatory Requirements

- This system will guarantee privacy of user information.
- The information of the user will not be distributed to third parties.
- The system will meet minimum standards of data protection and privacy of web-based systems.

5.6 Reuse Objectives

- The application will also be made in a modular way to enable other parts of the system to be reused like authentication, user management, and booking modules.
- Reusable components can also be reused or incorporated in the future in the works that are related to sports management or online booking system.

6. Revised Project Plan

This part will provide the latest advancement of the Sports Ecosystem Platform and a new project plan in accordance to the initial proposal. The plan provides details of activities done, work done and work to be done. The schedule of the project and task dependencies are represented in a Gantt chart.

6.1 Project Progress Status

- **Completed**.
    - Requirement elicitation
    - Software purpose The Software requirements specification (SRS).
    - High-level system design
    - Table identification AKA schema identification.
- **In Progress**
    - Implementation front end user interface.
    - Backend API development
    - Database integration
- **Planned**
    - System validation and testing.
    - fixing of bugs and optimization.
    - Last documentation and deployment.
    - Best presentation and defense preparation of project. (Sommerville, 2015)

6.2 Updated Project Plan (Phases)

Phase	Activities	Duration
Phase 1	Requirement Analysis & SRS	2 Weeks
Phase 2	System Design & Modeling	1 Week
Phase 3	Database Design	1 Week
Phase 4	Frontend Development	3 Weeks
Phase 5	Backend Development	3 Weeks
Phase 6	Integration & Testing	2 Weeks
Phase 7	Final Deployment & Documentation	1 Week

1. References

[1] Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach* (9th Edition). McGraw-Hill Education, New York. Pages 145-189 (Understanding Requirements). ISBN: 978-1260548006. Available at: https://www.mheducation.com/highered/product/software-engineering-practitioner-s-approach-pressman-maxim/M9781260548006.html

[2] Dennis, A., Wixom, B. H., & Tegarden, D. (2021). *Systems Analysis and Design: An Object-Oriented Approach with UML* (6th Edition). John Wiley & Sons, Hoboken, New Jersey. Pages 87-134 (Requirements Determination). ISBN: 978-1119803782. Available at: https://www.wiley.com/en-us/Systems+Analysis+and+Design%3A+An+Object+Oriented+Approach+with+UML%2C+6th+Edition-p-9781119803782

[3] Pohl, K., & Rupp, C. (2021). *Requirements Engineering Fundamentals: A Study Guide for the Certified Professional for Requirements Engineering Exam* (3rd Edition). Rocky Nook, San Rafael, California. Pages 112-167 (Requirements Elicitation and Documentation). ISBN: 978-1681988825. Available at: https://rockynook.com/shop/photography/requirements-engineering-fundamentals-3rd-edition/

[4] IEEE Computer Society. (2018). *ISO/IEC/IEEE International Standard - Systems and Software Engineering - Life Cycle Processes - Requirements Engineering*. IEEE Std 29148-2018. Institute of Electrical and Electronics Engineers, New York. Pages 1-104. Available at: https://standards.ieee.org/standard/29148-2018.html

[5] Leffingwell, D., & Widrig, D. (2020). *Managing Software Requirements: A Use Case Approach* (3rd Edition). Addison-Wesley Professional, Boston. Pages 201-256 (Use Case Modeling). ISBN: 978-0135792667. Available at: https://www.pearson.com/en-us/subject-catalog/p/managing-software-requirements-a-use-case-approach/P200000003289

[6] Seidl, M., Scholz, M., Huemer, C., & Kappel, G. (2022). *UML @ Classroom: An Introduction to Object-Oriented Modeling* (2nd Edition). Springer International Publishing, Cham, Switzerland. Pages 45-98 (UML Diagrams and Notation). ISBN: 978-3030828448. DOI: 10.1007/978-3-030-82845-5. Available at: https://link.springer.com/book/10.1007/978-3-030-82845-5

[7] Ramakrishnan, R., & Gehrke, J. (2023). *Database Management Systems* (4th Edition). McGraw-Hill Education, New York. Pages 34-89 (Entity-Relationship Model and Database Design). ISBN: 978-1260507904. Available at: https://www.mheducation.com/highered/product/database-management-systems-ramakrishnan-gehrke/M9781260507904.html

[8] Kumar, A., Singh, R., & Sharma, P. (2022). Web-Based Sports Management Systems: A Comprehensive Review. *International Journal of Advanced Computer Science and Applications (IJACSA)*, Volume 13, Issue 6. Pages 234-242. DOI: 10.14569/IJACSA.2022.0130629. Available at: https://thesai.org/Publications/ViewPaper?Volume=13&Issue=6&Code=IJACSA&SerialNo=29

Appendix A: Glossary

A. Definitions

Term	Definition
Admin	System administrator responsible for verification, monitoring, approvals, and system management.
Player	Registered user who can book training sessions, reserve grounds, buy equipment, and track performance.
Coach	Verified professional who provides training, manages schedules, and evaluates player performance.
Business Owner	User who manages an online sports equipment store on the platform.
Attendance Tracking	Recording player presence or absence in training sessions.
Coach Recommendation System	Automated feature that suggests suitable coaches based on player data like sport, skill level, and location.
CRUD Operations	Basic database operations: Create, Read, Update, Delete.
Dashboard	Main interface showing summary, notifications, and system features for each user role.
Ground Booking	Reservation of indoor sports facilities for specific dates and times.
Performance Points	Scores assigned by coaches to evaluate player skills and progress.
Role-Based Access Control (RBAC)	Security system that restricts access based on user role.
Subscription Package	Paid plan allowing Business Owners to list products (Basic, Pro, Premium).
Training Plan	Weekly schedule of exercises and goals assigned by coaches.
Training Session	Scheduled practice meeting between coach and player.
Verification	Admin approval process for coach and business owner documents.
Listing Slot	Maximum number of products allowed under a subscription package
Low Stock Alert	Notification when product inventory is low.

Performance Dashboard	Visual interface showing graphs of player progress.
Pending Status	Account state before admin approval.

B. Acronyms and Abbreviations

Acronym	Full Form
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DBMS	Database Management System
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
JSON	JavaScript Object Notation
PDF	Portable Document Format
RBAC	Role-Based Access Control
SRS	Software Requirements Specification
SQL	Structured Query Language
UML	Unified Modeling Language
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator

C.Project-Specific Terms

Term	Definition
Automated Coach Matching	System logic that matches players with coaches automatically.
Indoor Ground	Sports facility available for booking (Cricket or Badminton).
Admin Panel	Control panel used by admin to manage users and system.
Notification System	Feature that sends alerts about bookings, payments, and approvals.
Subscription Renewal	Process of extending business owner listing package after expiry.

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
…					

Table 1: List of non-trivial defects